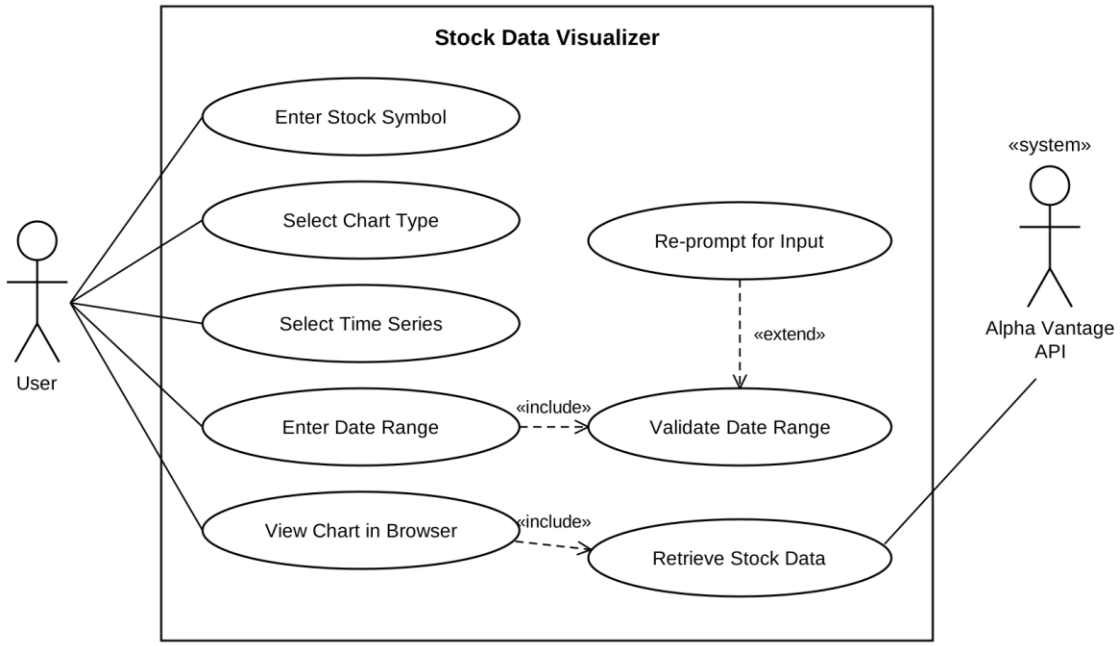
Use Case Diagram



The User is the primary actor and goes through each step of the program. Alpha Vantage is a secondary actor because the program depends on it for the stock data but it never starts anything on its own. Enter Date Range always includes Validate Date Range, and View Chart in Browser always includes Retrieve Stock Data. Re-prompt for Input extends Validate Date Range since it only happens when a date fails the checks.

Use Case Descriptions

UC-1: Enter Stock Symbol

Use Case Name	Enter Stock Symbol
Use Case ID	UC-1
Primary Actor	User
Secondary Actor	None
Description	The user types in the ticker symbol for the company they want to see stock data for.
Trigger	The user starts the program.
Preconditions	The program is running and has shown the first prompt.
Main Flow	1. The system asks the user to enter a stock symbol. 2. The user types a symbol like AAPL or MSFT and presses Enter. 3. The system changes the input to uppercase and removes any extra spaces. 4. The system saves the symbol to use in the API request later.
Alternate Flows	a. If the user leaves the input blank the system says the symbol cannot be empty and asks again (UC-6). b. If the symbol is not a real company the program does not know yet. This gets caught later when the API sends back an error instead of data (UC-7).
Postconditions	A stock symbol is saved and the program moves on to the chart type prompt.

UC-2: Select Chart Type

Use Case Name	Select Chart Type
Use Case ID	UC-2
Primary Actor	User
Secondary Actor	None
Description	The user picks what kind of chart they want the stock data shown in.
Trigger	A stock symbol has been entered.
Preconditions	UC-1 is finished and a symbol is saved.

Main Flow	1. The system shows a numbered list of chart types (1. Bar 2. Line). 2. The user types the number of the chart they want. 3. The system checks that the number is one of the options. 4. The system saves the chart type.
Alternate Flows	If the user types a number that is not on the list, a letter or nothing at all, the system says the choice is not valid and shows the list again (UC-6).
Postconditions	A valid chart type is saved and the program moves on to the time series prompt.

UC-3: Select Time Series

Use Case Name	Select Time Series
Use Case ID	UC-3
Primary Actor	User
Secondary Actor	None
Description	The user picks which Alpha Vantage time series function the program uses, which decides how far apart the data points are.
Trigger	A chart type has been selected.
Preconditions	UC-2 is finished and a chart type is saved.
Main Flow	1. The system shows a numbered list of time series options (1. Intraday 2. Daily 3. Weekly 4. Monthly). 2. The user types the number of the option they want. 3. The system checks that the number is one of the options. 4. The system matches the choice to the right API function name, for example 2 becomes TIME_SERIES_DAILY.
Alternate Flows	If the input is not one of the listed numbers the system says the choice is not valid and shows the list again (UC-6).
Postconditions	The API function name is saved for the request.

UC-4: Enter Date Range

Use Case Name	Enter Date Range
Use Case ID	UC-4
Primary Actor	User

Secondary Actor	None
Description	The user enters the start date and end date for the stock data they want to look at.
Trigger	A time series function has been selected.
Preconditions	UC-3 is finished and the API function is saved.
Main Flow	1. The system asks for the start date in YYYY-MM-DD format. 2. The user types the start date. 3. The system checks the start date (UC-5). 4. The system asks for the end date in YYYY-MM-DD format. 5. The user types the end date. 6. The system checks the end date and makes sure it is not before the start date (UC-5).
Alternate Flows	If either date fails the checks in UC-5 the system tells the user what was wrong and asks for that date again (UC-6).
Postconditions	A valid start date and end date are saved.

UC-5: Validate Date Range

Use Case Name	Validate Date Range
Use Case ID	UC-5
Primary Actor	System
Secondary Actor	None
Description	The system makes sure both dates are real dates in the right format and that the range makes sense. This is included every time the user enters a date in UC-4.
Trigger	The user enters a start date or an end date.
Preconditions	The user has typed something at a date prompt.
Main Flow	1. The system tries to read the input as a YYYY-MM-DD date. 2. The system checks that it is a real calendar date (for example 2024-02-30 would fail). 3. Once both dates are entered the system checks that the end date is the same as or after the start date. 4. If everything passes the dates are accepted.

Alternate Flows	a. If the format is wrong (like 10/23/2024) the system says the date has to be in YYYY-MM-DD format (UC-6). b. If the date does not exist on the calendar the system says it is not a valid date (UC-6). c. If the end date is before the start date the system says the end date cannot be before the start date and asks for the end date again (UC-6).
Postconditions	The date range is confirmed valid and the program continues.

UC-6: Re-prompt for Input

Use Case Name	Re-prompt for Input
Use Case ID	UC-6
Primary Actor	System
Secondary Actor	None
Description	When the user enters something the program cannot use, the system explains the problem and asks again instead of crashing. This extends UC-5 and is also used by the menus in UC-1 through UC-3.
Trigger	An input fails validation.
Preconditions	The user entered an input that was rejected.
Main Flow	1. The system shows a short message that says what was wrong with the input. 2. The system shows the same prompt again. 3. The user enters a new value. 4. The system checks the new value. This repeats until the input is valid.
Alternate Flows	None. The loop only ends once a valid input is entered.
Postconditions	The program is still running and the user was able to fix their mistake.

UC-7: Retrieve Stock Data

Use Case Name	Retrieve Stock Data
Use Case ID	UC-7
Primary Actor	System
Secondary Actor	Alpha Vantage API

Description	The system sends a request to the Alpha Vantage API with the user's choices and then keeps only the data that falls inside the date range. This is included in UC-8.
Trigger	All of the user's inputs have been entered and checked.
Preconditions	UC-1 through UC-5 are finished and a valid API key is set up in the program.
Main Flow	1. The system builds the request URL using the time series function, stock symbol and API key. 2. The system sends the request to Alpha Vantage. 3. Alpha Vantage sends back the stock data in JSON format. 4. The system goes through the data and keeps only the dates between the start date and end date. 5. The system pulls out the open, high, low and close prices for each of those dates.
Alternate Flows	a. If the API sends back an error message (bad symbol or the daily request limit was reached) the system shows the message to the user and does not try to make a chart. b. If the computer has no internet connection the request error is caught and the system tells the user it could not reach Alpha Vantage. c. If no data falls inside the date range (for example the range is on a weekend or before the company went public) the system tells the user and asks them to pick a different range.
Postconditions	The filtered stock data is ready to be put in a chart.

UC-8: View Chart in Browser

Use Case Name	View Chart in Browser
Use Case ID	UC-8
Primary Actor	User
Secondary Actor	None
Description	The system makes the chart from the stock data and opens it in the user's default web browser.
Trigger	The user finishes entering all of the inputs.
Preconditions	UC-7 returned data inside the date range.

Main Flow	1. The system runs UC-7 to get the stock data. 2. The system makes the chart using the chart type the user picked, with dates along the x axis and the open, high, low and close prices as separate lines or bars. 3. The system gives the chart a title with the stock symbol and the date range. 4. The system opens the chart in the user's default browser. 5. The system asks if the user wants to look at another stock.
Alternate Flows	a. If UC-7 fails no chart is made and the user sees the error message instead. b. If the user says yes at step 5 the program goes back to UC-1. If they say no the program ends.
Postconditions	The user is looking at a chart of the stock data they asked for.